An exceedingly brief introduction to GitLab and the cluster

Hey look we have a GitLab isn't that cool oh also a cluster neat

Lukas Burk

Leibniz Institute for Prevention Research and Epidemiology — BIPS

2026-11-02

Jour Fixe

`git` and GitLab

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project
- Can by synchronized with a **server**, such as...

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else
  (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project
- Can by synchronized with a **server**, such as...

GitLab

- GitLab.com $\approx$ GitHub.com alternative
- Private BIPS server at `srvgit.bips.eu`

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project
- Can by synchronized with a **server**, such as...

GitLab

- GitLab.com $\approx$ GitHub.com alternative
- Private BIPS server at `srvgit.bips.eu`
- Supplements existing (public) use of GitHub.com

What is `git` and GitLab

`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project
- Can by synchronized with a **server**, such as...

GitLab

- GitLab.com $\approx$ GitHub.com alternative
- Private BIPS server at `srvgit.bips.eu`
- Supplements existing (public) use of GitHub.com
- Hosts `git` repositories: typically code / analyses

What is `git` and GitLab

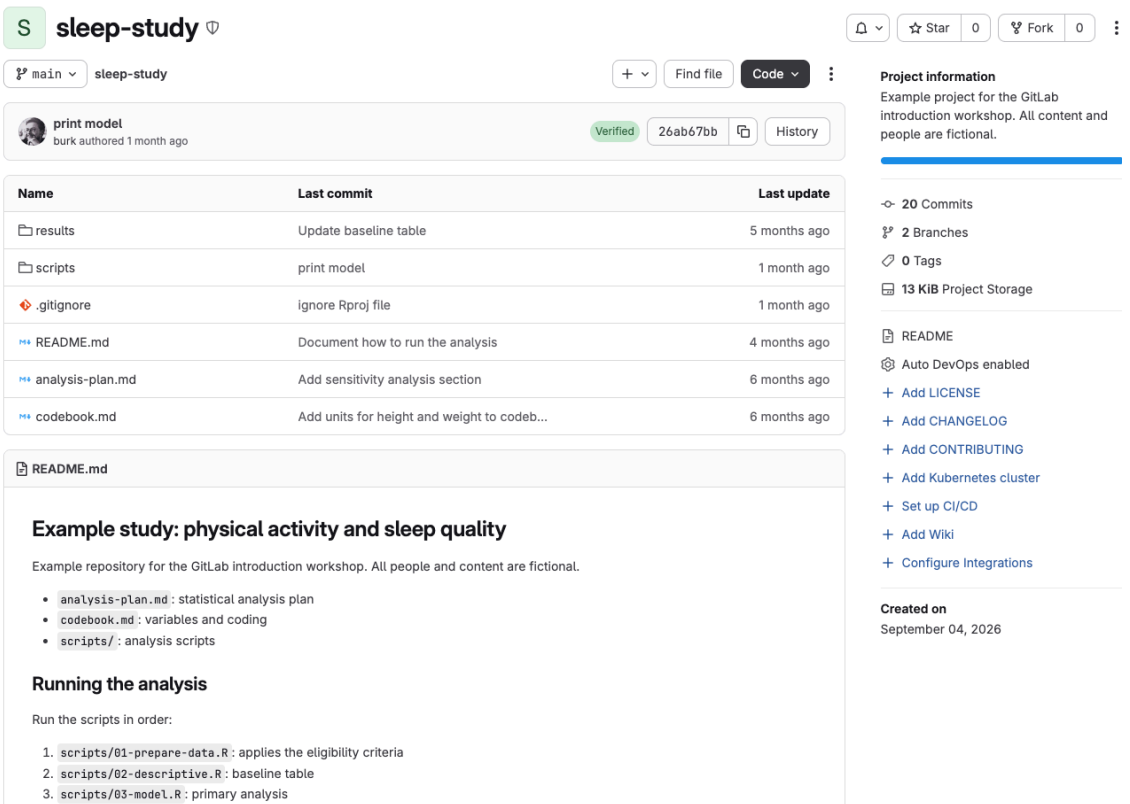
`git`

- Version control program for **code** / plain text
- Standard development practice
- Installed on your computer and wherever else (workstations, cluster, termserv)
- A folder using `git` is a **repository**
  - Saves the changes between code versions
  - Each “checkpoint” is a **commit**
  - Retains **complete history** of a project
- Can by synchronized with a **server**, such as...

GitLab

- GitLab.com ≈ GitHub.com alternative
- Private BIPS server at `srvgit.bips.eu`
- Supplements existing (public) use of GitHub.com
- Hosts `git` repositories: typically code / analyses
- Adds: **Project management and collaboration**
  - issue boards (e.g., todo list), discussions
  - Work on same code (merge requests)
  - `@`-tagging users you collaborate with
- View results if produced in RMarkdown/Quarto!

GitLab: Example repository



S **sleep-study**

Star 0 | Fork 0

main | sleep-study | Find file | Code

print model
burk authored 1 month ago | Verified | 26ab67bb | History

Name	Last commit	Last update
results	Update baseline table	5 months ago
scripts	print model	1 month ago
.gitignore	ignore Rproj file	1 month ago
README.md	Document how to run the analysis	4 months ago
analysis-plan.md	Add sensitivity analysis section	6 months ago
codebook.md	Add units for height and weight to codeb...	6 months ago

README.md

Example study: physical activity and sleep quality

Example repository for the GitLab introduction workshop. All people and content are fictional.

- `analysis-plan.md`: statistical analysis plan
- `codebook.md`: variables and coding
- `scripts/`: analysis scripts

Running the analysis

Run the scripts in order:

1. `scripts/01-prepare-data.R`: applies the eligibility criteria
2. `scripts/02-descriptive.R`: baseline table
3. `scripts/03-model.R`: primary analysis

Project information
Example project for the GitLab introduction workshop. All content and people are fictional.

20 Commits
2 Branches
0 Tags
13 KiB Project Storage

README
Auto DevOps enabled
+ Add LICENSE
+ Add CHANGELOG
+ Add CONTRIBUTING
+ Add Kubernetes cluster
+ Set up CI/CD
+ Add Wiki
+ Configure Integrations

Created on
September 04, 2026

GitLab: Per-line code history

01-prepare-data.R 381 B

months ago	Add data preparation sc...	1
		2
		3
		4
		5
		6
		7
month ago	fix inclusion criteria (clo...	8
		9
months ago	Add data preparation sc...	10
		11
		12
		13
		14
		15
		16
		17

Authored 1 month ago

fix inclusion criteria (closes #5)

burk bd8b8e3b

```
# Build the analysis dataset by applying the eligibility criteria
library(dplyr)

raw <- readr::read_csv("data/raw/enrolment.csv")

analysis_data <- raw |>
  filter(
    age >= 40,
    age <= 65,
    consent == "yes",
    !apnoea,
    !shift_work_12m
  ) |>
  select(id, age, sex, pa_min, psqi, n_missing_psqi, height, weigh

readr::write_csv(analysis_data, "data/analysis.csv")
```

Where repositories live

- Users belong to **groups**, which share access
- Repositories should be *created under a group*
- Everyone here: Member of the **BIPS group**

Where repositories live

- Users belong to **groups**, which share access
- Repositories should be *created under a group*
- Everyone here: Member of the **BIPS group**
- → Central group ensures collaboration
- Groups can have nested subgroups
  - Entities like **NAKO**, **IDEFICS/I.Family**
  - Research units
  - Topic groups
  - ...

Where repositories live

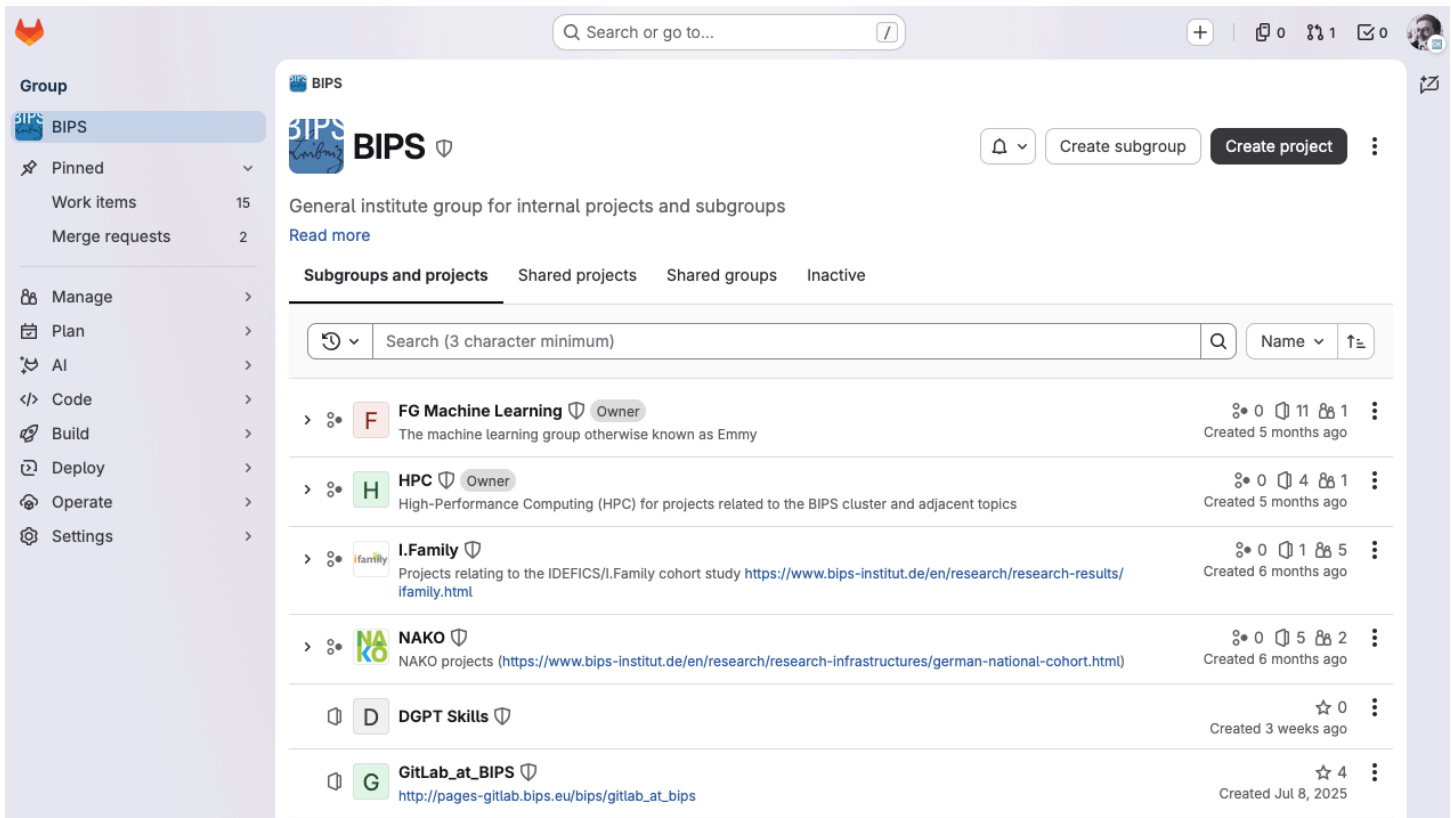
- Users belong to **groups**, which share access
- Repositories should be *created under a group*
- Everyone here: Member of the **BIPS group**
- → Central group ensures collaboration
- Groups can have nested subgroups
  - Entities like **NAKO**, **IDEFICS/I.Family**
  - Research units
  - Topic groups
  - …

> **i** We will flexibly adapt this going forward

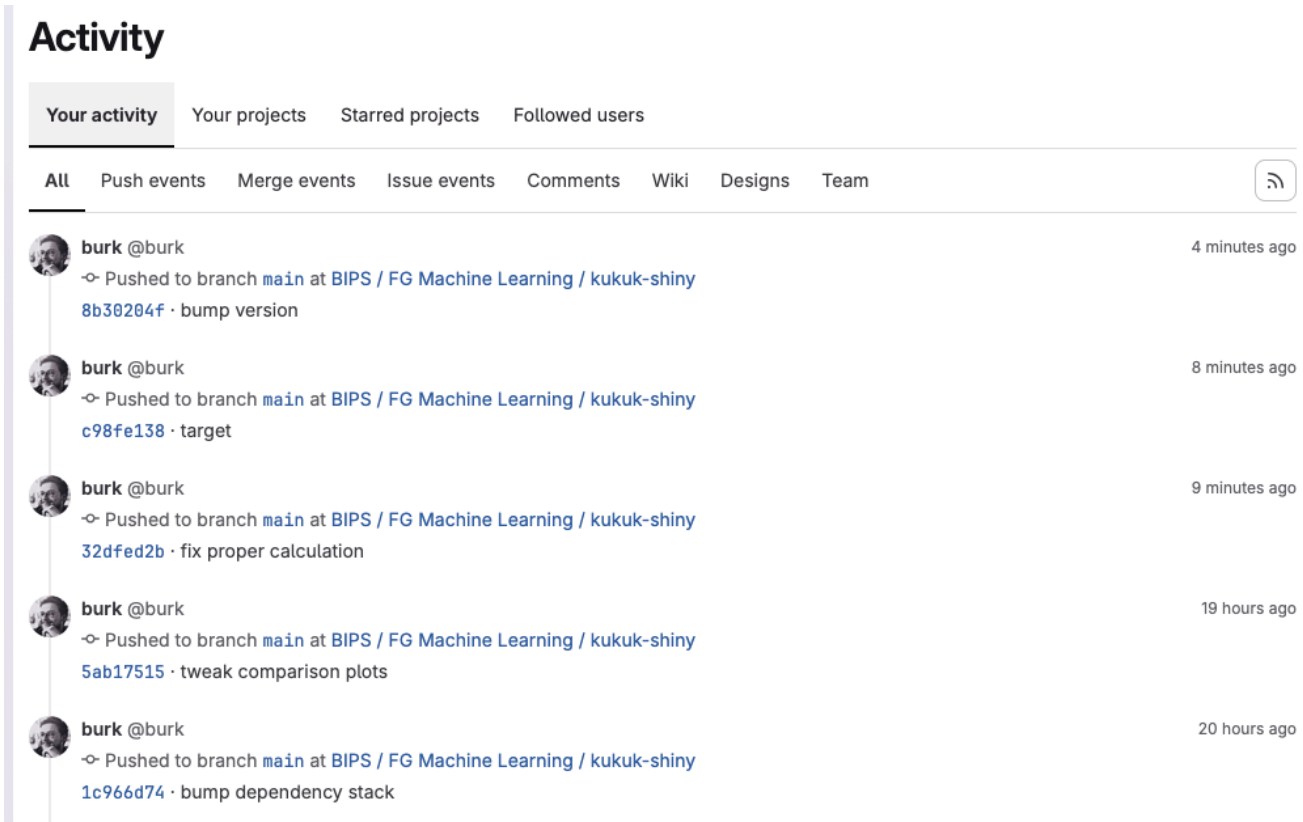
Example repository hierarchy

- BIPS
  - HPC
    - bips-cluster-user-docs
  - FG ML
    - kukuk-shiny
  - NAKO
    - …
  - IDEFCS/I.Family
    - …
- burk
  - gitlab-intro

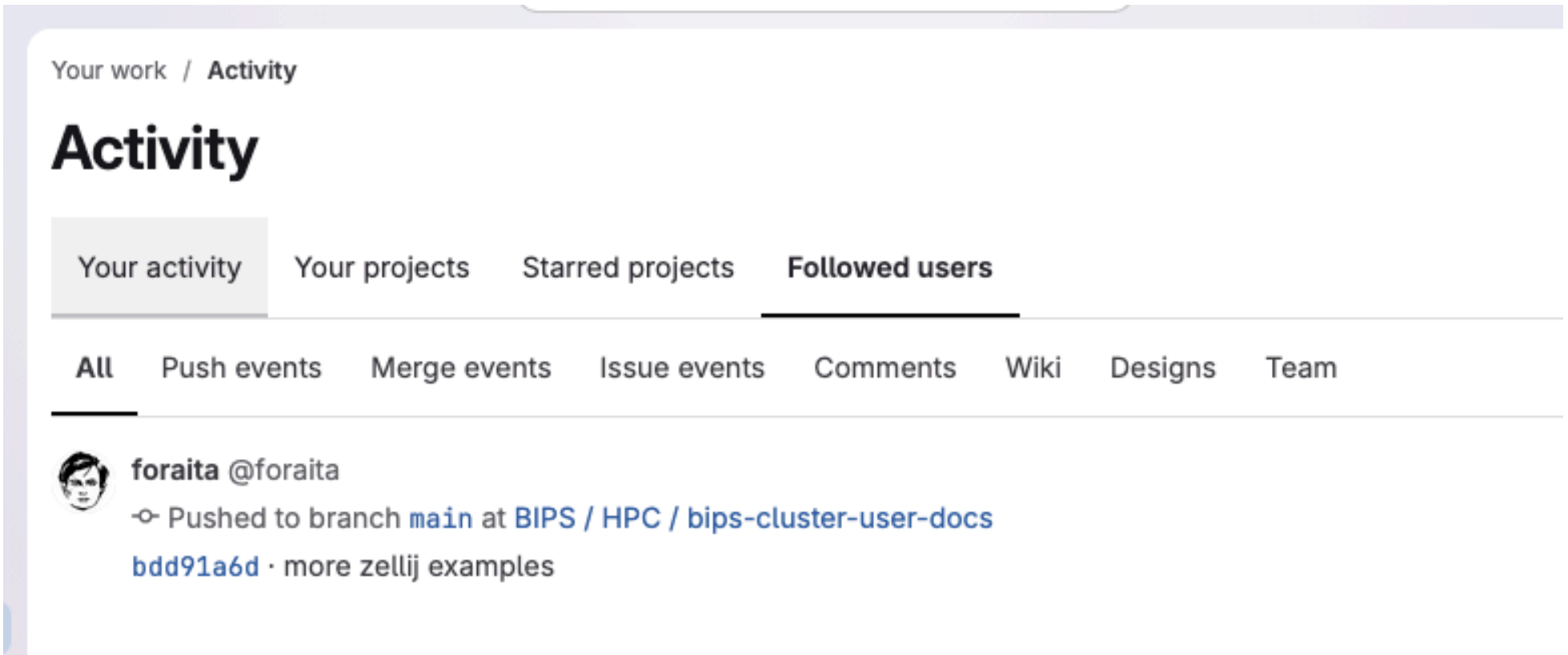
GitLab: BIPS Group



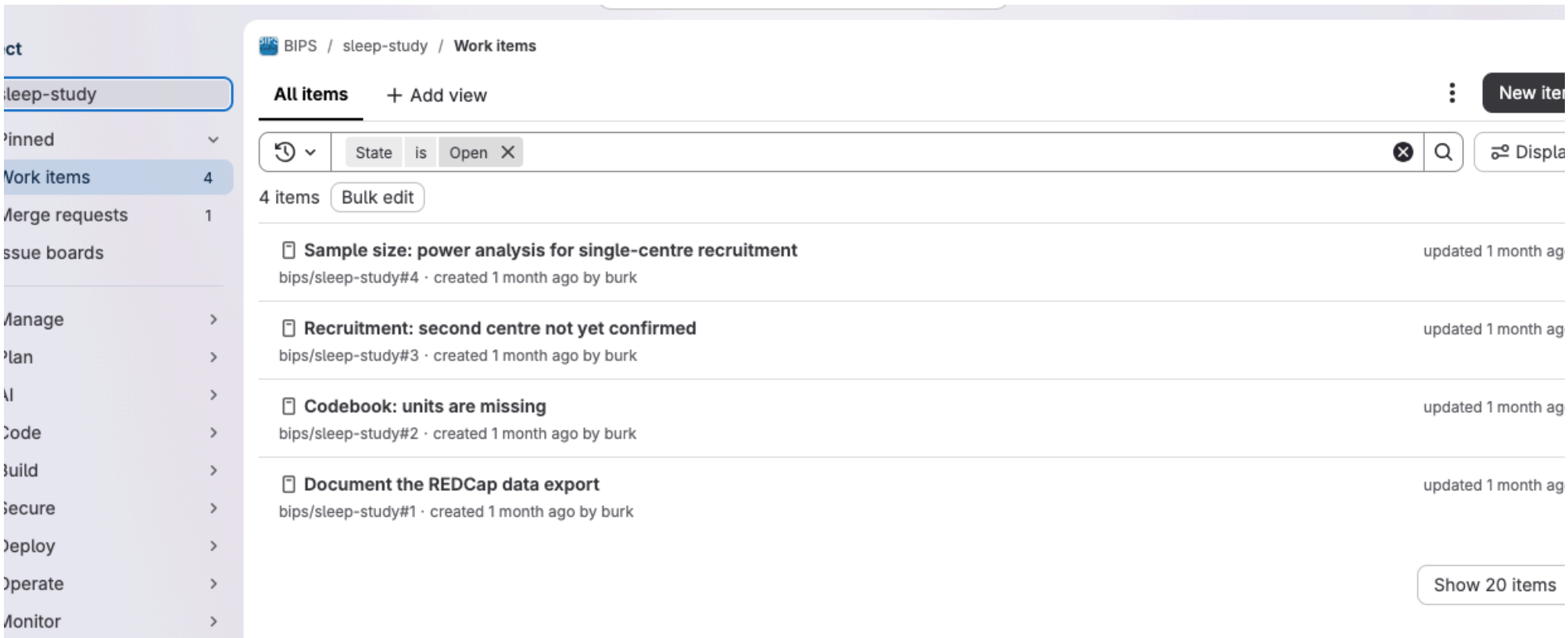
GitLab: Activity feed



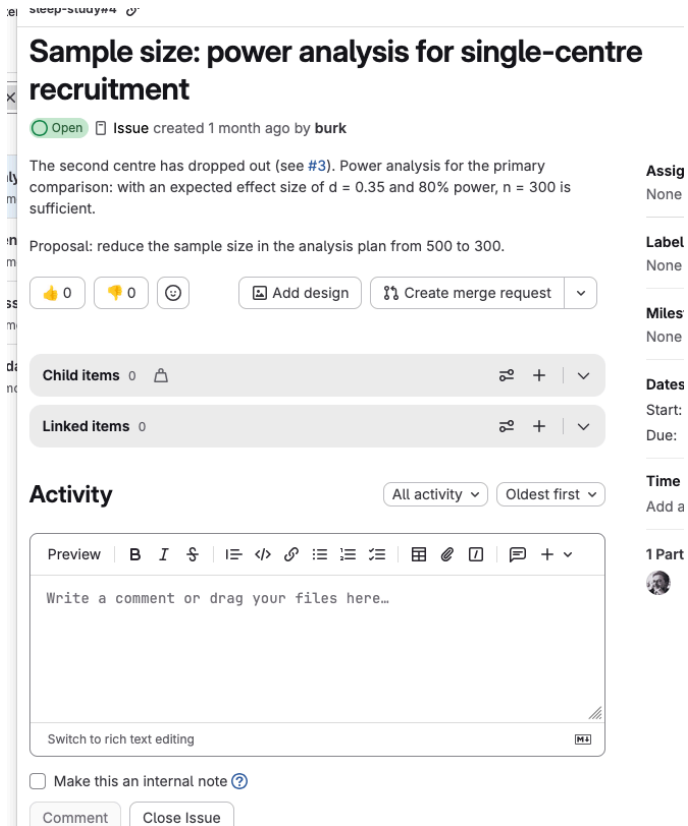
GitLab: Activity feed



GitLab: Project issues



GitLab: Project issues



GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**
- Trackable *who* changed *what*, *when*, and *why*
- Older versions can be restored and compared (thanks to `git`)

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**
- Trackable *who* changed *what*, *when*, and *why*
- Older versions can be restored and compared (thanks to `git`)
- Analysis results → RMarkdown/Quarto → viewable live (accessible also for non-coders)

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**
- Trackable *who* changed *what*, *when*, and *why*
- Older versions can be restored and compared (thanks to `git`)
- Analysis results → RMarkdown/Quarto → viewable live (accessible also for non-coders)

!! There will be trainings

Different scopes for PhDs, supervisors, programmers, …

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**
- Trackable *who* changed *what*, *when*, and *why*
- Older versions can be restored and compared (thanks to `git`)
- Analysis results → RMarkdown/Quarto → viewable live (accessible also for non-coders)

!! There will be trainings

Different scopes for PhDs, supervisors, programmers, …

Planned next steps will be *announced via e-mail*!

GitLab: We're still getting started!

What stays

- `G:\` remains generic file storage
- OX-Drive remains
- Word, Excel, PDF, datasets: unchanged
- Public R packages and paper supplements stay on *GitHub*

What changes

- Analysis code is `git`-versioned on GitLab rather than manually
- Collaboration via **issues** and **merge requests**
- Trackable *who* changed *what*, *when*, and *why*
- Older versions can be restored and compared (thanks to `git`)
- Analysis results → RMarkdown/Quarto → viewable live (accessible also for non-coders)

!! There will be trainings

Different scopes for PhDs, supervisors, programmers, …
Planned next steps will be *announced via e-mail*!
Optional: *Reach out to me* for dedicated sessions

The BIPS HPC cluster

What's in a cluster

Many ways to shave a yak, and now you have one more!

What's in a cluster

Many ways to shave a yak, and now you have one more!

Your PC/laptop

- You know it
- Shorter tasks
- Not so powerful
- Completely your own
- Convenience: 💅

What's in a cluster

Many ways to shave a yak, and now you have one more!

Your PC/laptop

- You know it
- Shorter tasks
- Not so powerful
- Completely your own
- Convenience: 💅

Server

- TermServ / workstations
- Run *looong* tasks
- 1 powerful computer
- *Shared*
- Convenience: Mixed

What's in a cluster

Many ways to shave a yak, and now you have one more!

Your PC/laptop

- You know it
- Shorter tasks
- Not so powerful
- Completely your own
- Convenience: 💅

Server

- TermServ / workstations
- Run *looong* tasks
- 1 powerful computer
- *Shared*
- Convenience: Mixed

HPC Cluster

- Many powerful computers connected
- >1000 CPU cores, 2 GPUs, terabytes of RAM
- Access via *head node*
- *Shared* (resources managed via scheduler)

What's in a cluster

Many ways to shave a yak, and now you have one more!

Your PC/laptop

- You know it
- Shorter tasks
- Not so powerful
- Completely your own
- Convenience: 💅

Server

- TermServ / workstations
- Run *looong* tasks
- 1 powerful computer
- *Shared*
- Convenience: Mixed

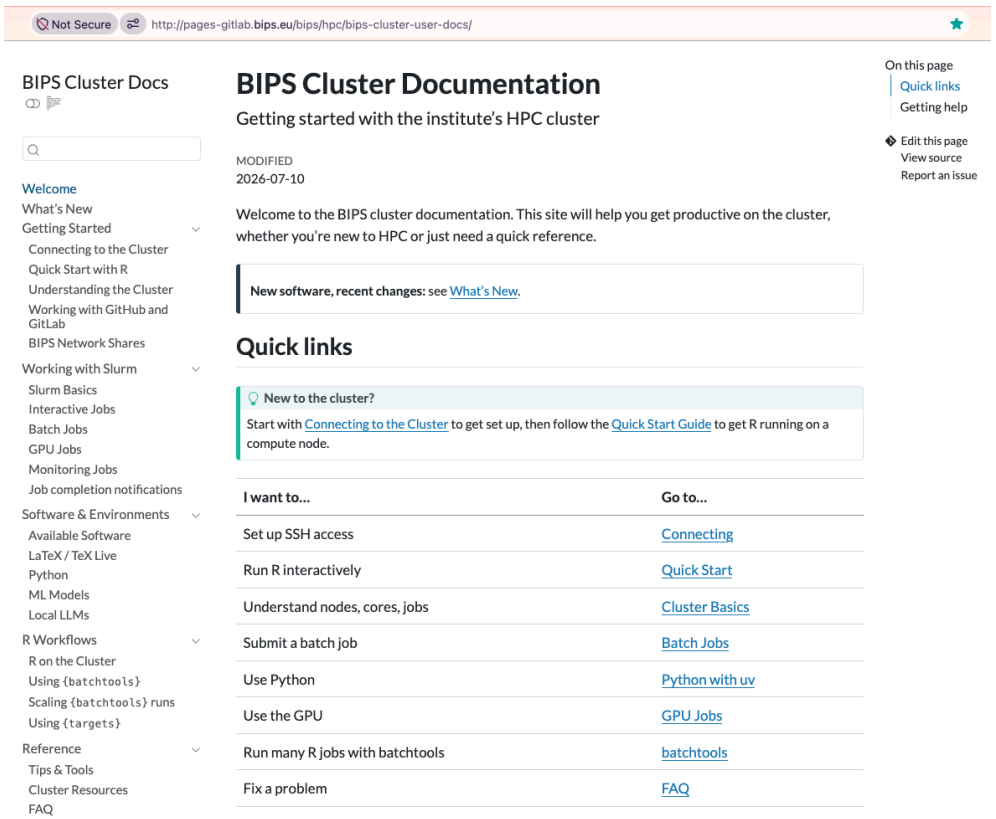
HPC Cluster

- Many powerful computers connected
- >1000 CPU cores, 2 GPUs, terabytes of RAM
- Access via *head node*
- *Shared* (resources managed via scheduler)
- Convenience:
  - Learning curve (Linux, Slurm, ...)
  - You'll get there

Cluster docs rendered via GitLab CI

Deployed to GitLab Pages, viewable in institute network

http://pages-gitlab.bips.eu/bips/hpc/bips-cluster-user-docs



Cluster

Get access & learn more

1. *Ask me for an account*, I will send you login credentials (`burk`)
2. Bookmark the dashboard (current cluster usage):
   - http://srvcluster.bips.de/ (no https!)
3. Read (and bookmark) the docs:
   - http://pages-gitlab.bips.eu/bips/hpc/bips-cluster-user-docs
4. Try demos / usage examples for `batchtools`, and others on GitLab:
   - http://srvgit.bips.eu/bips/bips-cluster-demos

Thank you for your attention!

Contact

Lukas Burk
Leibniz Institute for Prevention Research and Epidemiology — BIPS
Achterstraße 30
28359 Bremen
Germany

burk@leibniz-bips.de